

IIFS: An improved incremental feature selection method for protein sequence processing

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ABSTRACT

Motivation: Discrete features can be obtained from protein sequences using a feature extraction method. These features are the basis of downstream processing of protein data, but it is necessary to screen and select some important features from them as they generally have data redundancy.

Result: Here, we report IIFS, an improved incremental feature selection method that exploits a new subset search strategy to find the optimal feature set. IIFS combines nonadjacent sorting features to prevent the drawbacks of data explosion and excessive reliance on feature sorting results. The comparative experimental results on 27 feature sorting data show that IIFS can find more accurate and important features compared to existing methods. The IIFS approach also handles data redundancy more efficiently and finds more representative and discriminatory features while ensuring minimal feature dimensionality and good evaluation metrics. Moreover, we wrap this method and deploy it on a web server for access at <http://112.124.26.17:8005/>.

1. Introduction

Feature selection methods play an important role in protein sequence processing [1–3]. In feature selection, relevant features associated with the target task are retained, while irrelevant or redundant features are removed to constitute the optimal feature set. This optimal feature set can be applied to construct efficient classification models in protein recognition [4–9]. Feature selection plays a crucial role in protein sequence processing because it can filter out more representative and discriminative features, thus improving the performance and accuracy of the model. In an ideal situation, it is necessary to traverse all feature combinations from the selected feature set to generate an optimal candidate feature subset. However, with an increase in the dimension of the feature set to be selected, the feature selection method leads to the combinatorial explosion problem [10–15]. It also tends to suffer from high computational complexity and tendency to fall into local optimization. Generally, feature selection is based on the result of feature sorting and selects an optimal feature subset. The feature selection

procedure consists of two parts: feature subset search and subset evaluation. Based on whether the classifier to be used is introduced in the subset evaluation process, it is further divided into filter-based and wrap-based methods; mRMR [16] and Relief belong to the former, while ANOVA [17] and MRMD [18] are often combined with the final classifier to form wrapped feature selection methods. The importance of a feature in mRMR is determined by its correlation with other remaining features and sample labels. If the redundancy between other features is low and the correlation with sample labels is high, then this feature is considered important. In addition, the subset search usually adopts heuristic algorithms. The statistical measure of the importance of a feature portrayed by relief through “near-hit” and “near-miss”, similar to mRMR, also requires a threshold to select the optimal feature set. The wrapped feature selection method not only includes the selected classifier but also takes the sorting result of the feature sorting algorithm as the input. For example, MRMD series algorithms (MRMD1.0, MRMD2.0 and MRMD3.0) [19,20] and ANOVA are often used as the feature sorting part of the wrapped feature selection algorithm. The input of the

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wrapped feature selection method is the features obtained by the feature extraction method or a combination of multiple features; it sorts the features into the selected classifier through a forward search strategy, backward search strategy or a hybrid mode. All feature selection methods use the performance of the classifier as the evaluation index to determine the dimension of the optimal feature set.

It has been found that filtered feature selection may include empirical or human factors, while wrapped feature selection overly relies on the sorting results of selected features. Therefore, in the present work, we use feature subset search as a breakthrough point and propose an algorithm based on an elimination strategy to address the above issues.

2. Materials and methods

2.1. Benchmark dataset

Three types of protein sequence data are used in the comparative experiment. They are obtained in a similar way but there are some differences in the specific parameter settings and screening process. Specifically, the benchmark dataset used to identify polystyrene binding peptides includes 104 positive samples (polystyrene binding peptides) and 104 negative samples (nonpolystyrene binding peptides) [21]. The raw data for identifying thermophilic proteins is Lin's dataset (915 thermophilic proteins and 793 non thermophilic proteins) [22–24]. The data used for cancerlectin protein recognition are from the Cancer-ElectroninDB dataset constructed by Damodaran et al., which includes 178 positive samples and 226 negative samples [25,26]. The dataset was collected from the PDB database (released in January 2017) according to the following criteria. Raw positive samples were selected from nine different phage display libraries. In the order of presentation, the above three datasets are represented below as 'peptide', 'thermophilic' and 'cancer'.

2.2. Feature extraction

We randomly combine the above three basic datasets with several feature extraction methods and obtain 9 feature datasets. These feature extraction algorithms all use protein sequences as input to extract structural, physicochemical or statistical information, which are discrete features that include information such as hydrophilicity, hydrophobicity or amino acid occurrence frequency [27]. Specifically, the 473D feature extraction method includes evolutionary information and secondary structure information of protein sequences. A total of 473 dimensional features were extracted through PSI-BLAST and PSI-PRED, with 440 features obtained through PSI-BLAST and 33 dimensional features obtained through PSI-PRED. The 400D feature is the frequency characteristic of the appearance of dipeptides in the protein sequence, while the 420D feature is the frequency characteristic of a single amino acid residue with 20 dimensions combined with the above 400D feature. 188D and iztli(27D) extracted a series of physicochemical feature information of protein sequences.

2.3. Feature sorting

To compare and verify the effectiveness of the proposed algorithm, we use the ANOVA feature sorting algorithm, the Euler distance metric mrm_d_e feature sorting algorithm and the MRMD feature sorting algorithm on the above 9 feature datasets and obtain 27 sorted feature datasets. These 27 sorted feature datasets are the final dataset for verifying the effectiveness of the proposed method.

2.4. Classification method

The identification of a specific type of protein sequence in the above 27 datasets is essentially a classification problem. In protein sequence identification, traditional machine learning methods such as naive Bayes

[28,29], logical regression [30,31], random forest [32,33], decision tree [34] and support vector machine [35–37] can be used as classification methods, and deep learning networks [38–42] can also be used to build classifiers. However, in the construction of protein sequence recognizers, support vector machines [43], as a classic binary classification algorithm, have been widely applied and achieved good results. By mapping the sample points to high-dimensional space, the algorithm constructs a hyperplane to maximize the distance between samples and make them linearly separable.

2.5. Evaluation metrics

Due to the balance and imbalance in the above dataset, accuracy, sensitivity, specificity, F1 score and AUC are used as evaluation metrics [44,45]. Their definitions are as follows.

$$ACC = \frac{TN + TP}{TN + FP + FN + TP} \quad (1)$$

$$SN = \frac{TP}{TP + FN} \quad (2)$$

$$SP = \frac{TN}{TN + FP} \quad (3)$$

$$AUC = P(P_{positive} > P_{negative}) \quad (4)$$

$$F1_{score} = \frac{2TP}{2TP + FP + FN} \quad (5)$$

TN, TP, FN, and FP represent the number of true positives, true negatives, false-positives, and false-negatives, respectively. SN and SP represent sensitivity and specificity, respectively, reflecting the degree of classification of positive and negative samples. AUC is a specialized indicator used to evaluate the classification performance of binary classification models. The F1 score can intuitively reflect the model classification level of positive samples.

3. Results and discussion

3.1. Algorithm overview

The core of this algorithm is to maintain two linked list node structures. The first is used to store the recognition performance evaluation value and corresponding feature sequence of the feature combination calculated up to the current feature. The second is used to calculate the new feature sequence recognition performance evaluation value and corresponding feature sequence of the combination between the next feature and the feature sequence maintained in the first linked list. Whether to eliminate the next feature by comparing the recognition performance evaluation indicators in the two linked list nodes mentioned above is determined. When the value of the second linked list node is greater than the value of the first linked list node, the second linked list node is used to replace the first linked list node, and vice versa, it is directly eliminated. The above process is iterated until all the features are calculated. Finally, the highest value of the evaluation index value and the corresponding feature combination are saved in the first linked list node, which is the optimal feature set required for model training. The principle is shown in Fig. 1. Specifically, ① the sorted features are obtained through data preprocessing of protein sequence data, followed by feature extraction and sorting. ② The classification evaluation value for step i is calculated by feeding the i -th feature and the current feature combination to a specific classifier. ③ The classification evaluation value of the $i+1$ step concatenates the $i+1$ st feature with the feature set obtained in the i step and calculates it through the specific classifier. ④ When the evaluation value of step $i+1$ is greater than step i , the feature set obtained by step $i+1$ is considered the optimal feature set, and vice versa; they are directly eliminated, and the next

iteration process begins. As described in the figure, when the evaluation index calculation result of step $i+1$ is 0.83, which is greater than the evaluation index 0.72 of step i , the corresponding feature $i+1$ will be added to the optimal feature set. If the evaluation value calculation result of step $i+1$ is 0.69, which is less than the evaluation index 0.72 of step i , then feature $i+1$ is directly eliminated, and the next iteration process begins. The iterative process runs until all features are calculated once.

3.2. Algorithm process description

The pseudocode for the IIFS algorithm process is as follows.

Among them, $value(N_1)$ and $sequence(N_1)$ return the evaluation index value and sequence of the linked list node N_i , where N_i represents the i th linked list node. This algorithm continues the process of traversing the entire feature sorting sequence of the IFS algorithm and maintains two linked list node structures to store evaluation indicators and sequence information up to the current feature and the next feature. The main body of this program process is a loop structure.

3.3. Result data analysis

The IFS method was experimentally compared with our proposed IIFS. The experimental comparison of the four characteristic data of polystyrene binding peptides is shown in Table 1. When the feature

IIFS Algorithm

Input Sorted feature data sequence $Fea'_n = [fea'_1, fea'_2, \dots, fea'_i, \dots, fea'_{n-1}, fea'_n]$

Output Best Feature Collection φ

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1 Applying for two linked list nodes in memory  $N_1, N_2$ 
2 for  $i \leftarrow 1$  to  $n$  do
3   if ( $value(N_1) == \text{null}$ ) then
4      $Fea'_i \rightarrow$  Classification Algorithm
5      $Fea'_i$  Evaluating Indicator  $\rightarrow N_1$ 
6      $Fea'_i \rightarrow N_1$ 
7   end if
8    $sequence(N_1) + fea'_{i+1} \rightarrow$  Classification Algorithm
9    $sequence(N_1) + fea'_{i+1}$  Evaluating Indicator  $\rightarrow N_2$ 
10   $N_1 + fea'_{i+1} \rightarrow N_2$ 
11  if  $value(N_1) < value(N_2)$  then
12     $N_1 \rightarrow N_2$ 
13  end if
14 end for
15 return Feature set  $sequence(N_1)$ 

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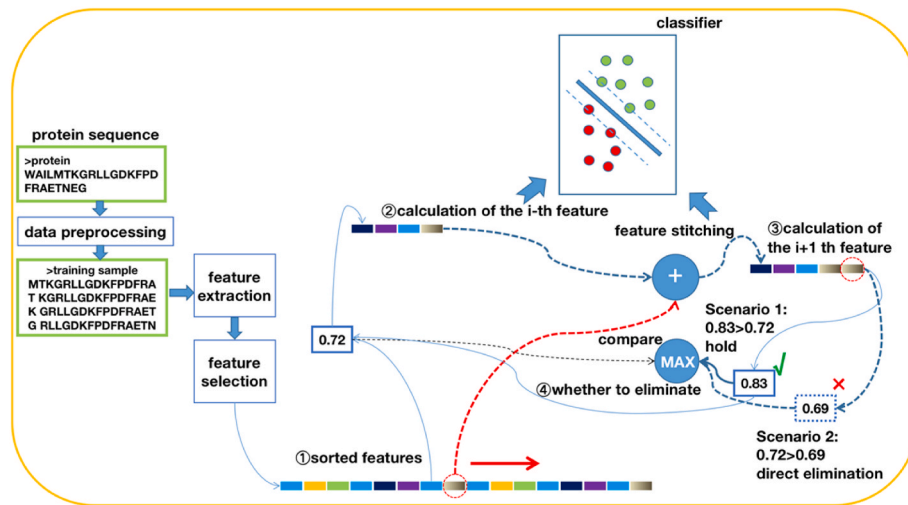


Fig. 1. Principle of the IIFS algorithm. The whole process is divided into four steps: ① preprocessing, feature extraction, and sorting of protein sequence data. ② Calculate the performance of the i -th feature combined with the current feature set. ③ Calculate the connection between the $i+1$ feature and the feature set of ②. ④ If the evaluation value of ③ is greater than ②, the $i+1$ feature will be retained; otherwise, it will be directly eliminated.

Table 1

Subset search on polystyrene binding peptide feature data.

Dataset	Feature sorting	Feature selection	Dimension	ACC%	SN	SP	F1	AUC
peptide_188D	ANOVA	IIFS	9	70.1923	0.731	0.673	0.702	0.702
		IFS	55	71.1538	0.683	0.74	0.711	0.712
	mrmd_e	IIFS	15	63.4615	0.596	0.673	0.634	0.635
		IFS	113	68.75	0.721	0.654	0.687	0.688
	MRMD	IIFS	14	68.2692	0.75	0.615	0.681	0.683
		IFS	105	71.1538	0.702	0.721	0.712	0.712
peptide_400D	ANOVA	IIFS	18	75.4808	0.74	0.769	0.755	0.755
		IFS	179	76.9231	0.769	0.769	0.769	0.769
	mrmd_e	IIFS	23	61.0577	0.519	0.702	0.607	0.611
		IFS	117	79.8077	0.798	0.798	0.798	0.798
	MRMD	IIFS	26	64.4231	0.606	0.683	0.644	0.644
		IFS	158	67.3077	0.654	0.692	0.673	0.673
peptide_420D	ANOVA	IIFS	17	74.5192	0.74	0.75	0.745	0.745
		IFS	179	76.9231	0.769	0.769	0.769	0.769
	mrmd_e	IIFS	23	61.0577	0.519	0.702	0.607	0.611
		IFS	117	79.8077	0.798	0.798	0.798	0.798
	MRMD	IIFS	121	62.0192	0.471	0.769	0.612	0.62
		IFS	177	63.9423	0.712	0.567	0.638	0.639
peptide_iztli	ANOVA	IIFS	6	75	0.673	0.827	0.749	0.75
		IFS	7	75.9615	0.683	0.837	0.758	0.76
	mrmd_e	IIFS	9	69.7115	0.702	0.692	0.697	0.697
		IFS	20	72.5962	0.683	0.769	0.725	0.726
	MRMD	IIFS	6	70.6731	0.644	0.769	0.706	0.707
		IFS	11	72.1154	0.673	0.769	0.721	0.721

extraction method, feature sorting method, and feature subset search strategy are the peptide_400D feature, ANOVA, IFS, or IIFS methods, the ACC, SN, SP, F1, and AUC of the optimal feature set obtained are 76.9231%, 0.769, 0.769, 0.769, and 0.769, respectively, with a corresponding dimension of 179. The ACC, SN, SP, F1, and AUC of the sub-optimal combination are 75.4808%, 0.74, 0.769, 0.755, and 0.755, respectively. The difference between it and the optimal combination is that the IIFS method is used, and the dimension of the optimal feature set is 18.

Therefore, both combinations mentioned above can be used to train recognition models for polystyrene binding peptides. Their ACC, SN, SP, F1, and AUC differ by only 1.4423%, 0.029, 0, 0.014, and 0.014, respectively, while their dimensions differ by 161 dimensions. Therefore, the experimental results indicate that these statistical features have a relatively small contribution rate in recognition. Among the 12 feature sorting data generated by polystyrene bound peptides, the IIFS method achieved similar performance to IFS in all 10 groups, with ACC differences within 3%. The dimension of the optimal feature set

corresponding to IIFS is much smaller than that of the original IFS algorithm, with an average difference of approximately 4 times.

The comparison of the four subset search algorithms on thermophilic protein feature sorting data is shown in Table 2. From the perspective of five evaluation indicators, ACC, SN, SP, F1, and AUC, it was found that all combinations in the table achieved good results, all above 0.88. This result indicates that the Thermophilic_188D and Thermophilic_420D features contain key features for identifying thermophilic proteins. However, the dimensionality of the optimal feature set corresponding to the original IFS is relatively high, so the combination obtained through this subset search algorithm is not suitable for being selected as an option for constructing recognition models. The minimum dimension of the optimal feature set is 22, which is the result of using the IIFS method on ANOVA sorting of Thermophilic_188D feature data. The evaluation index values of ACC, SN, SP, F1, and AUC are 92.2581%, 0.926, 0.919, 0.923, and 0.922, respectively. Therefore, a new optimal feature set with lower dimensions and evaluation indicators within a reasonable range should be selected as the identification model option.

Table 2
Subset search on thermophilic protein feature data.

Dataset	Feature sorting	Feature selection	Dimension	ACC%	SN	SP	F1	AUC
Thermophilic_188D	ANOVA	IIFS	22	91.5543	0.914	0.918	0.916	0.916
		IFS	56	92.2581	0.926	0.919	0.923	0.922
	mrm_d_e	IIFS	35	89.912	0.902	0.896	0.899	0.899
		IFS	163	90.5572	0.907	0.904	0.906	0.905
	MRMD	IIFS	34	90.6745	0.903	0.912	0.907	0.907
		IFS	137	91.1437	0.916	0.906	0.911	0.911
Thermophilic_420D	ANOVA	IIFS	26	90.8504	0.912	0.904	0.909	0.908
		IFS	228	92.434	0.909	0.942	0.924	0.926
	mrm_d_e	IIFS	32	89.4428	0.895	0.894	0.894	0.894
		IFS	217	91.7889	0.905	0.933	0.918	0.919
	MRMD	IIFS	47	88.8563	0.904	0.871	0.888	0.887
		IFS	191	90.6745	0.9	0.914	0.907	0.907

Table 3 shows the comparison of four subset search algorithms on cancer cell junction protein feature sorting data. After analyzing all combinations in the table from the perspectives of five evaluation indicators, including ACC, SN, SP, F1, and AUC, it was found that no good results were obtained, which means they cannot be directly used to construct recognition models. This result indicates that the cancer_188D feature, cancer_400D feature and cancer_473D feature may not contain the important key features for identifying cancer cell junction proteins. However, this experiment is mainly used to compare the performance of subset search algorithms. When the feature extraction method, feature sorting method, and feature subset search strategy are cancer_473D features, ANOVA, and IIFS methods, the ACC, SN, SP, F1, and AUC obtained are 68.0693%, 0.556, 0.779, 0.676, and 0.667, respectively. The dimension of its corresponding optimal feature set is 19.

Overall, the dimension of the optimal feature set corresponding to IIFS is smaller than that of the original IFS algorithm, and the difference in the five evaluation indicators is relatively small. Therefore, it can be considered that compared with the original IFS subset search algorithm, the IIFS algorithm based on the elimination strategy avoids the combination explosion caused by the latter's mechanical increase in the number of features to obtain the optimal evaluation index by combining nonadjacent features with eliminated features.

3.4. Process data analysis

We analyzed the process data and compared the differences between our proposed IIFS algorithm and the original IFS algorithm in process data processing. We visualized the processing process of the IIFS algorithm and original IFS algorithm on the 188D features of polystyrene bound peptides, ANOVA and mrm_d_e sorting data of 27-dimensional iztli features. The results are shown in Fig. 2, where the horizontal

axis represents the sorted features, and the vertical axis represents the evaluation indicators corresponding to the candidate subset. Here, the ACC value is selected. This figure shows the comparison results of the sorting results of ANOVA and mrm_d_e of the 188D feature of polystyrene binding peptide and the 27-dimensional iztli feature, named after the combination of feature and sorting method. For example, 188D_ANOVA refers to the sorting results of features obtained by using the sorting algorithm of the ANOVA feature on the 188D feature. Similarly, the processing process of the above two subset search algorithms is visualized on the ANOVA and MRMD sorting data of 473D characteristics of cancer cell junction proteins and 420D characteristics of thermophilic proteins, named 473D_ANOVA, 473D_MRMD, 420D_ANOVA and 420D_MRMD, respectively, as shown in Fig. 3.

In comparison, the IFS algorithm achieves good evaluation metrics while ensuring the minimum feature dimension. The original IFS algorithm selects the optimal feature set with a higher dimension and exhibits the so-called "V" phenomenon, where the evaluation metrics fluctuate up and down as the number of features increases. The ACC value of IIFS only increases as the number of features increases. Specifically, on iztli_ANOVA, either due to iztli being a candidate feature with only 27 dimensions or due to the reasonable order of sorting features, the ACC values of traditional IFS do not fluctuate on this dataset. On the remaining seven comparison results, the ACC values of the original IFS algorithm not only fluctuate up and down but also the dimensionality of the selected optimal feature set is much higher than that of IIFS. Further analysis revealed that the subset search process of the original IFS overly relied on feature sorting results, each time adding features to search for feature combinations that added ACC, while the IIFS algorithm based on the elimination strategy found higher ACC values by combining nonadjacent features. The adjacent nodes in the line chart corresponding to the IIFS span several features, while the

Table 3
Subset search on cancer cell junction protein feature data.

Dataset	Feature sorting	Feature selection	Dimension	ACC%	SN	SP	F1	AUC
cancer_188D	ANOVA	IIFS	7	65.8416	0.545	0.748	0.655	0.646
		IFS	18	65.8416	0.545	0.748	0.655	0.646
	mrm_d_e	IIFS	22	63.1188	0.433	0.788	0.618	0.61
		IFS	108	63.8614	0.562	0.699	0.637	0.63
	MRMD	IIFS	14	56.4356	0.017	0.996	0.417	0.506
		IFS	85	61.8812	0.438	0.761	0.608	0.6
cancer_400D	ANOVA	IIFS	6	65.099	0.506	0.765	0.645	0.636
		IFS	34	66.8317	0.635	0.695	0.669	0.665
	mrm_d_e	IIFS	12	65.8416	0.483	0.796	0.649	0.64
		IFS	85	65.8416	0.579	0.721	0.657	0.65
	MRMD	IIFS	4	59.1584	0.191	0.907	0.527	0.549
		IFS	165	63.8614	0.579	0.686	0.638	0.632
cancer_473D	ANOVA	IIFS	19	68.0693	0.556	0.779	0.676	0.667
		IFS	64	70.0495	0.573	0.801	0.696	0.687
	mrm_d_e	IIFS	19	64.3564	0.461	0.788	0.633	0.624
		IFS	134	68.8119	0.584	0.77	0.685	0.677
	MRMD	IIFS	9	55.9406	0	1	?	0.5
		IFS	427	58.9109	0.455	0.695	0.584	0.575

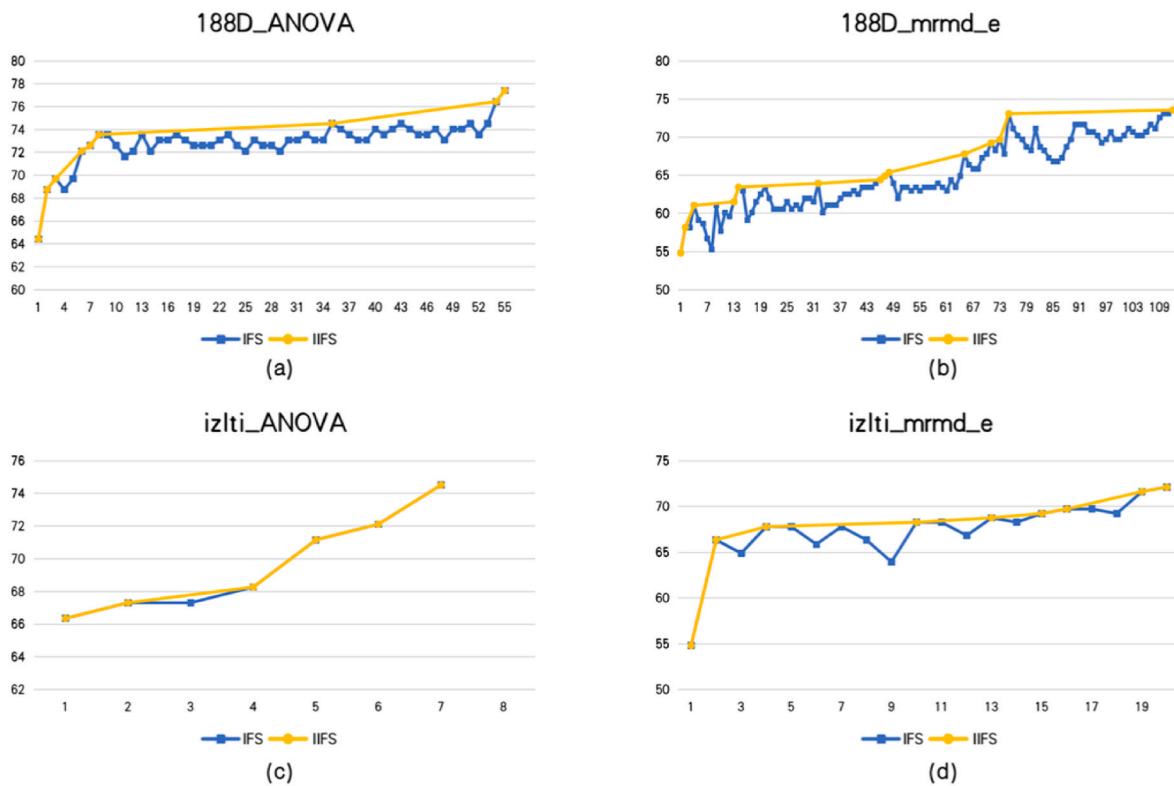


Fig. 2. Sort the 188D features and 27-dimensional izlti features of polystyrene bound peptides using ANOVA and mrmd_e, respectively. The visualization results using the IFS and IIFS seed set search algorithms are shown in Fig. 2. (a), (b), (c), and (d), where the horizontal axis represents the sorted features and the vertical axis represents the ACC values corresponding to the candidate subset.

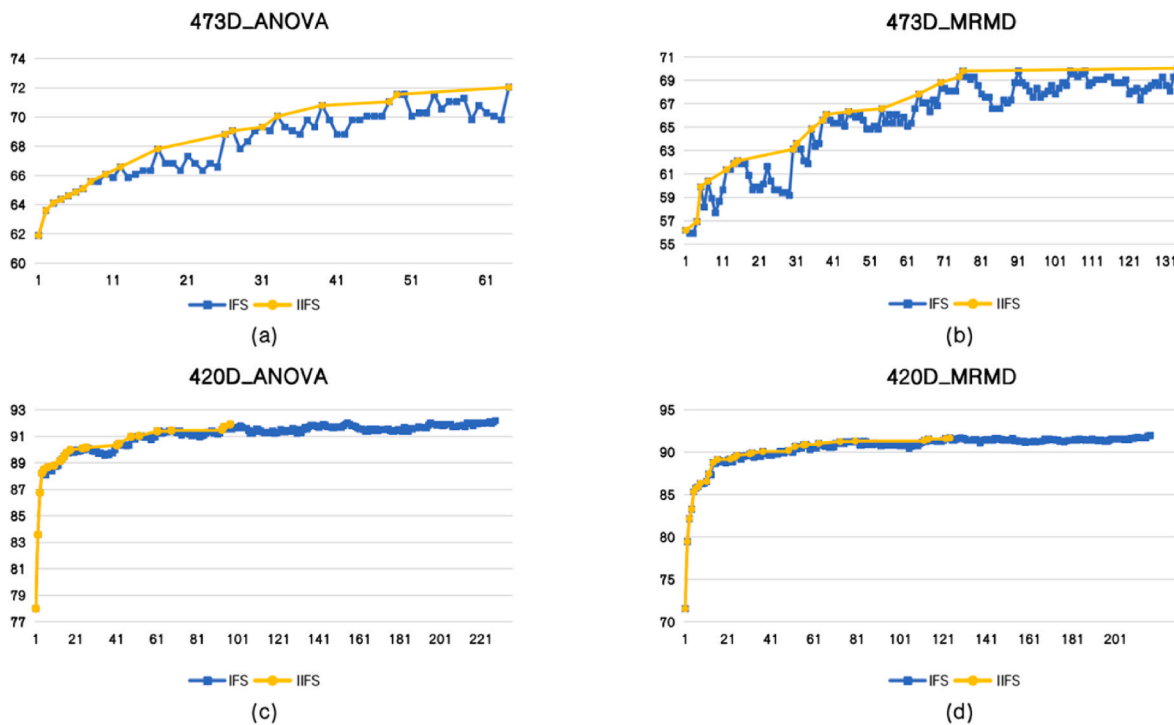


Fig. 3. The 473D and 420D features of polystyrene binding peptides were sorted by ANOVA and MRMD, respectively. The visualization results using the IFS and IIFS seed set search algorithms are shown in Fig. 3. (a), (b), (c), and (d), where the horizontal axis represents the sorted features, and the vertical axis represents the ACC value corresponding to the candidate subset.

original IFS selects the adjacent features as the elements of the optimal feature set. When the IIFS algorithm increases the evaluation index value by combining nonadjacent features, the original IFS can only find the optimal combination of adjacent features. In summary, the evaluation indicators of the IIFS algorithm for the proposed elimination strategy effectively avoid the “V” phenomenon in the original IFS algorithm due to its internal design and do not overly rely on a certain feature sorting method.

4. Conclusion

In this paper, we propose a feature subset search algorithm based on an elimination strategy, which belongs to the forward subset search algorithm in wrapped feature selection. The purpose of this work is to discover the optimal feature set through the combination of nonadjacent features and ensure that the model evaluation index values corresponding to the feature traversal process will only increase and will not fluctuate up or down, as in the original IFS algorithm. Experiments on 21 feature sorting datasets of 3 protein sequences showed that the proposed feature subset search algorithm based on the elimination strategy outperformed the original IFS algorithm. Compared with the original IFS method, the IIFS method is more accurate and robust in selecting features, and can eliminate redundant features more effectively. Moreover, IIFS combines non-adjacent sorting features to avoid the drawbacks of the original IFS algorithm's data explosion and over-reliance on feature sorting results. IFS not only ensures advantages in evaluation indicators but also ensures that the selected optimal feature set has a smaller dimension than the original IFS algorithm. Therefore, the IFS algorithm based on the elimination strategy proposed in the process of generating the optimal feature subset is an effective supplement and adjustment to the upstream feature sorting results. The algorithm proposed in this paper has a balanced index and excellent performance, and in the future, we will consider to continue to modify the algorithm process so that it can be applied to more different protein identification work.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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